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Artificial intelligence assisted automated short answer question scoring tool shows high correlation with human examiner markings

H.M.T.W. Seneviratne^{1*} and S.S. Manathunga¹

Abstract

Background Optimizing the skill of answering Short answer questions (SAQ) in medical undergraduates with personalized feedback is challenging. With the increasing number of students and staff shortages this task is becoming practically difficult. Hence, we aimed to develop automated SAQ scoring tool (ASST) using artificial intelligence (AI) to evaluate written answers with feedback.

Methods This study investigated the use of large language models (LLMs) for automated SAQ scoring, specifically following rubrics. Rubric is a set of guidelines or criteria used to evaluate or grade assignments, which is provided by instructors. We focused on short answer questions from the Systematic Pharmacology course, along with model responses and rubrics shared with the LLM. The LLM analyzed student answers by extracting key parts, scoring them based on the rubric criteria, and providing feedback. The evaluation process relied on GPT-4, where the final score was determined by averaging results across five sampled runs. To validate the method, human examiners also graded the same answers, and the marks were compared to measure correlation.

Results Across 30 student answers, AES scores showed high correlation with independent human examiner markings (correlation coefficients of 0.93 and 0.96). An intra-class correlation coefficient of 0.94 indicate excellent inter-rater reliability between the LLM and human examiners.

Conclusion AI assisted automated SAQ scoring tool markings highly correlate with the human examiner markings which shows promise for transparent, flexible automated grading with granular feedback. This approach enabling criteria-based assessment could have valuable applications in education, reducing grading burdens on instructors while providing students with more granular, actionable feedback on their written work.

Keywords Automated scoring tool, Short answer questions, Artificial intelligence, Medical education

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Background

SAQs are widely used in undergraduate medical education for both summative and formative assessments. SAQs are designed to provide brief, precise responses to a specific question, encouraging clarity and focused thinking. SAQs can effectively assess a range of competencies, including higher-order cognitive skills [1]. However, providing personalized feedback on SAQ responses presents significant challenges, especially given increasing student numbers and limited faculty resources. Integrating AI-based automated scoring systems offers a promising solution—improving efficiency, ensuring consistent scoring, reducing bias, and saving valuable time [2].

Hence, we aim to develop an automated SAQ scoring tool powered by artificial intelligence (AI) to evaluate written responses while providing constructive feedback.

Automated Essay/SAQ Scoring leverages Natural Language Processing (NLP) techniques to evaluate written answers in academic settings. NLP is a form of artificial intelligence that allows computers to understand, interpret, and produce human language in a way that is both valuable and meaningful. The origins of these systems can be traced back to 1966 with the development of Project Essay Grade (PEG) by Ellis Batten Page. However, this early attempt was not sustainable at the time due to the prohibitive costs and limited computing power available [3]. After the 1990s, with the advent of more powerful desktop computers, several systems emerged to address the growing demand for efficient writing assessment tools. Notable examples include the Intelligent Essay Assessor (IEA) by Peter Foltz and Thomas Landauer, IntelliMetric by Vantage Learning, and the E-rater system developed by Educational Testing Service [4–6].

Early approaches primarily focused on developing mathematical models that related various features of essays to the final scores in a training dataset, using methods like linear regression. These models were then employed to score new essays within the same domain [7]. Subsequent systems utilized latent semantic analysis (LSA) to construct similarity measures between essays and reference materials [8]. However, a key limitation of these traditional models was their overly holistic nature, assigning a single score to the overall essay without granular feedback. To overcome this, later systems capable of assessing argument strength were developed by researchers like Isaac Persing and Vincent Ng, enabling evaluation of specific essay components [7]. This shift toward component-level assessment marked an important step in enhancing the diagnostic capabilities of Automated Scoring systems.

The scoring methodologies employed by Automated Essay/SAQ scoring systems can be broadly categorized into two aspects: technical feature analysis and semantic

analysis [9]. Technical features encompass numerical descriptors like word count, sentence count, and capitalization usage, which contribute to the final score. Semantic analysis, on the other hand, focuses on evaluating the meaning and conceptual coherence of sentences and paragraphs. Contemporary ASS systems often adopt a hybrid approach that assigns weighted scores to both technical features and semantic components. An example of this hybrid methodology is the use of Long Short-Term Memory (LSTM) networks combined with word embeddings for essay scoring, leveraging the strengths of both techniques [10].

Recent research has examined the use of LLMs in automated essay/SAQ scoring, producing mixed outcomes. While several studies underscore their advantages, others highlight key limitations. Comparative analyses of models such as ChatGPT, LLaMA, Google's PaLM 2, Anthropic's Claude 2, and OpenAI's GPT-3.5 and GPT-4 suggest that GPT-4 generally delivers superior performance [11]. In a cross-sectional study, Giacomo Rossetini and colleagues assessed the accuracy of ChatGPT-4, Microsoft Copilot, and Google Gemini on the Italian healthcare sciences entrance exam, finding that ChatGPT-4 and Microsoft Copilot outperformed Google Gemini [12].

However, these findings also emphasized the importance of carefully crafted prompts for achieving accurate and reliable assessments with LLMs [13]. Another study reported fluctuations in model performance over time, affecting interrater reliability [11]. Separately, Otaibi et al. developed a multi-agent system employing a Support Vector Machine (SVM) classifier to evaluate student exam questions based on Bloom's taxonomy [14].

While most existing automated essay/SAQ scoring models rely on the implicit construction of grading rubrics derived from training data, our study takes a fundamentally different approach. We investigate the use of pre-trained LLMs to develop an ASS system that scores SAQs based on an explicitly defined scoring rubric. Unlike conventional methods that require retraining on large, domain-specific essay corpora, our model leverages the inherent language understanding capabilities of LLMs to generate descriptive, transparent, and rubric-aligned evaluations—without the need for additional fine-tuning. This approach offers a more interpretable and adaptable alternative to traditional opaque AES systems, with the potential to improve clarity and reduce bias in grading.

The aims of the research:

Research Question:

Can a pre-trained large language model be used to accurately and transparently score short-answer questions, comparable to that of evaluations conducted by medical experts?

Methods

The study was conducted at the Department of Pharmacology, Faculty of Medicine, University of Peradeniya. Two SAQs were formulated from the domain of systematic pharmacology by the subject experts.

Domain experts were selected based on the qualifications and teaching experience. Comprehensive marking rubrics were developed by them, specifying the allocation of marks for each component of the ideal answer. These rubrics were further refined to include guidelines on assigning partial credit based on the quality and completeness of the response.

The questions were designed to suit third year medical undergraduates following pharmacology modules.

A LLM was aligned to assume the role of an expert examiner in systematic pharmacology. The original SAQ prompts, along with their respective marking rubrics and model answer keys, were incorporated into the system prompt during this alignment process. During inference, the student's response was provided as the user prompt.

Three consecutive prompt components were issued to the AI model: (1) to extract relevant excerpts from the student's answer that correspond to each component of the marking rubric (2), to assign marks based on the rubric and the extracted answer excerpts, and (3) to identify areas where the student demonstrated a lack of understanding and provide suggestions for improvement, such as restructuring the answer or addressing grammatical issues (Fig. 1).

To quantify the marking performance of the AI model, five independent assessments were generated for each student's answer. These assessments were obtained through temperature sampling from a truncated normal distribution centered at 0.3 with a standard deviation of 0.1, ensuring that the sampled temperatures remained within the range of 0 to 1. The underlying language model used for this process was GPT-4, a large language model trained by OpenAI [15]. This specific model was used because currently OpenAI GPT 4 iteration was ahead in most of reasoning benchmarks [11].

The median score across the five assessments was adopted as the final mark assigned by the AI model to the given student's answer. Median score was taken because the temperature sampling for the AI model does not guarantee a specific distribution for the final marks, so in case of skewed distributions, median is a more reasonable metric of central tendency, and in near normal case, it would be similar to the mean.

The variation in the assigned marks was then evaluated to assess the model's consistency and reliability in marking.

30 student responses (2 questions per student x 30 = 60 answers) were assessed using the ASS tool.

The same set of student answers was independently evaluated by two expert human examiners, who were tasked with marking the responses according to the established rubrics (Fig. 2).

Domain experts were selected based on the qualifications and experience in teaching and assessing pharmacology undergraduates.

Given the challenges in identifying suitably qualified subject matter experts, two examiners were selected based on their expertise in teaching and assessing undergraduate pharmacology students. Consequently, the evaluation was restricted to 30 student responses to ensure a feasible review process for the experts.

Although only a limited number of responses were evaluated, the post-hoc power analysis indicated a high statistical power, attributable to the large observed effect size despite the small sample size.

The relationship between total scores by the AI Marking tool and the two human evaluators were visualized using scatter plots and boxplots. Medians were compared with Kruskal-Wallis test. Pearson's product-moment correlation coefficients between AI and human examiner ratings were calculated. The reliability and consistency of the AI marking tool was evaluated using the intra-class correlation coefficient (ICC). A two-way mixed effects model (ICC3) was employed, treating subjects (students' answers) as a random effect and raters (the AI model and two human raters) as fixed effects [16]. A linear mixed-effects model with varying intercepts was fitted with following specifications.

$$score_i = \beta_0 + \beta_1 \times rater_i + \beta_{0i} + \epsilon_i$$

Where $score_i$ is the score for observation i , β_0 is the intercept, representing the overall mean score, β_1 is the coefficient for the fixed effect of "rater", representing the effect of the rater on the score, $rater_i$ is the rater for observation i , β_{0i} is the random intercept for subject i which is assumed to be normally distributed with mean 0 and variance $\sigma^2_{subject}$, and ϵ_i is the error term for observation i assumed to be normally distributed with mean 0 and variance σ^2 .

The variance components were extracted from the fitted model and the third-variation of ICC coefficient was calculated. Bland-Altman plots were generated to assess systematic bias in scoring the essay questions.

Results

The automated SAQ scoring performance of the developed LLM-based ASST was evaluated on a set of 30 student responses. Each essay question was scored out of 100. The final score was calculated as the average of the two scores and could range from 0 to 100%. The distributions of scores assigned by the AI marking tool and two

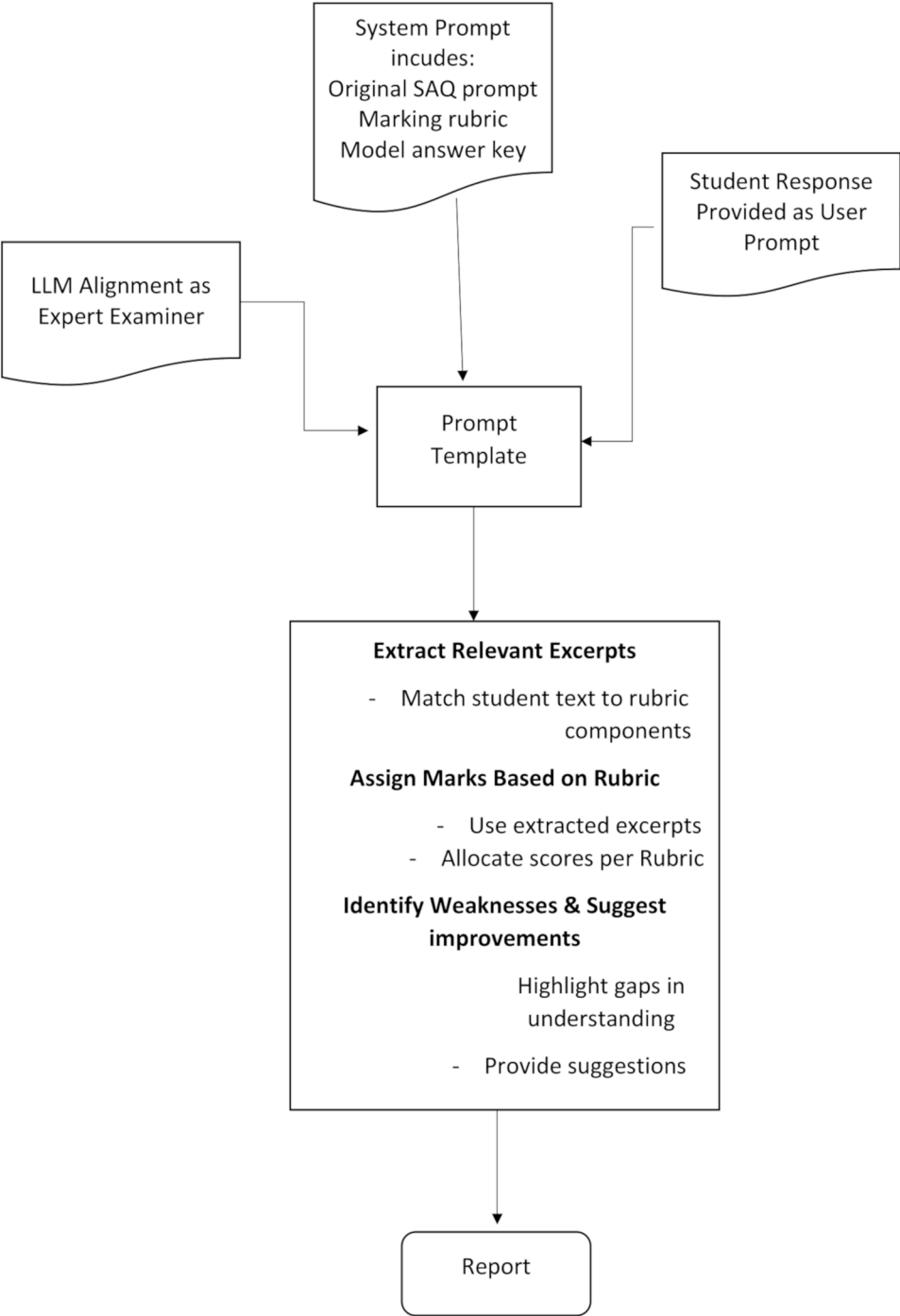


Fig. 1 Flow chart step 1: Designing the ASST

independent human examiners are visualized in Fig. 3. The Kruskal-Wallis test did not detect a statistically significant difference ($p=0.17$) between the median scores across the three raters.

The association between the AI score and the human evaluators’ score is depicted in the Fig. 4. The Pearson’s product moment correlation coefficient for AI and Examiner 1 was 0.93 ($p\text{-value}<0.001$,

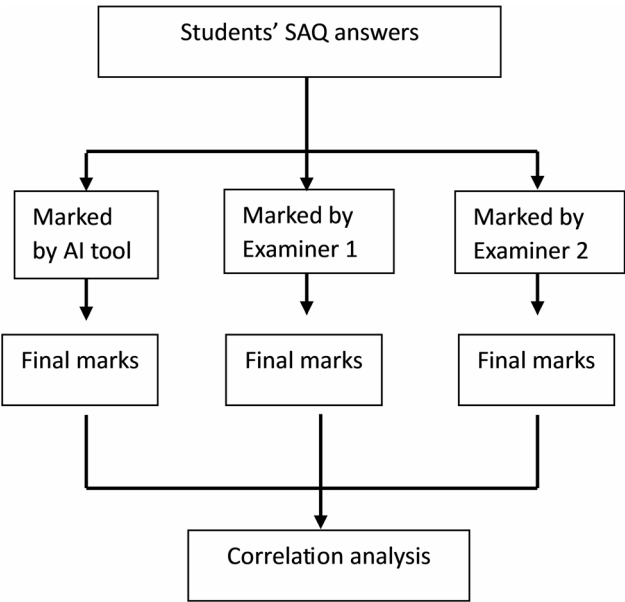


Fig. 2 Flow chart step 2 : Assessing the student answers

95% CI 0.85–0.96) and for AI and Examiner 2 was 0.96 (p -value < 0.001, 95% CI 0.90–0.98). The random-effect variance of the mixed effect model was 191.66 and the residual variance was 12.77. The ICC variation 3 coefficient estimate was 0.94 (p -value < 0.001, 95% CI 0.89–0.97) indicating an excellent inter-rater

reliability. The bias of scoring for the 1st human rater and the AI rater was 5.43 (95% CI 3.41–7.44) and the 2nd human rater and the AI rater was 0.96 (95% CI -0.61–2.54). The respective Bland-Altman plots are shown in Fig. 5.

Discussion

The results demonstrate that the proposed LLM-based approach shows promise for automatic SAQ scoring that aligns with specific rubrics provided by subject matter experts. The high correlation coefficients between the AI system’s scores and those of human examiners, along with the excellent inter-rater reliability measured by the ICC, suggest that the AI can evaluate SAQ’s in a manner that is consistent with expert grading.

A key advantage of this rubric-aligned LLM approach is the transparency and interpretability it provides. By explicitly incorporating the marking rubric into the prompt, the AI can generate scoring that closely follows the criteria outlined by instructors, rather than learning an opaque grading function from data. Moreover, since the AI does not require fine-tuning on a corpus of similar essays, this method could allow for more flexibility in scoring essays across diverse subject domains. As long as a comprehensive rubric is available, the AI could potentially assess responses in any topic area where it has

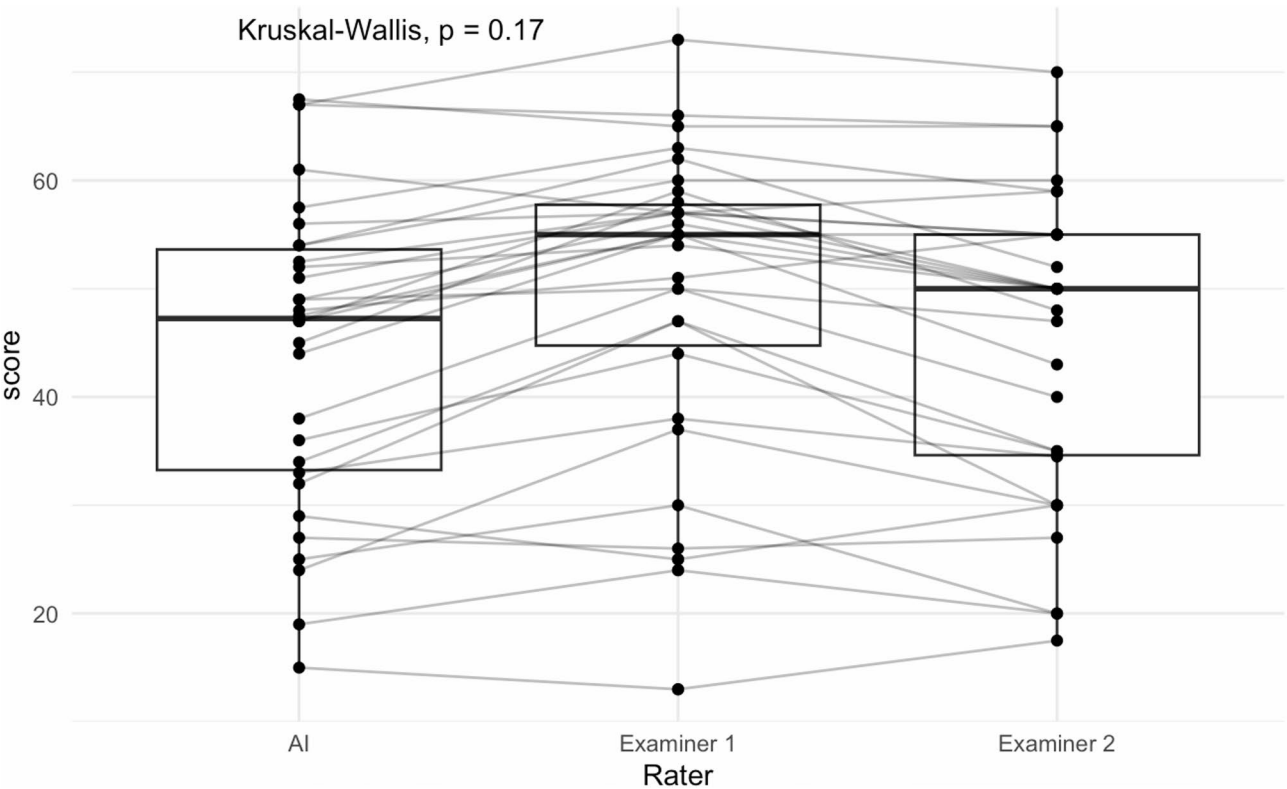


Fig. 3 The distribution of scores assigned by the AI marking tool (AI) and two independent human examiners (Examiner 1 and Examiner 2)

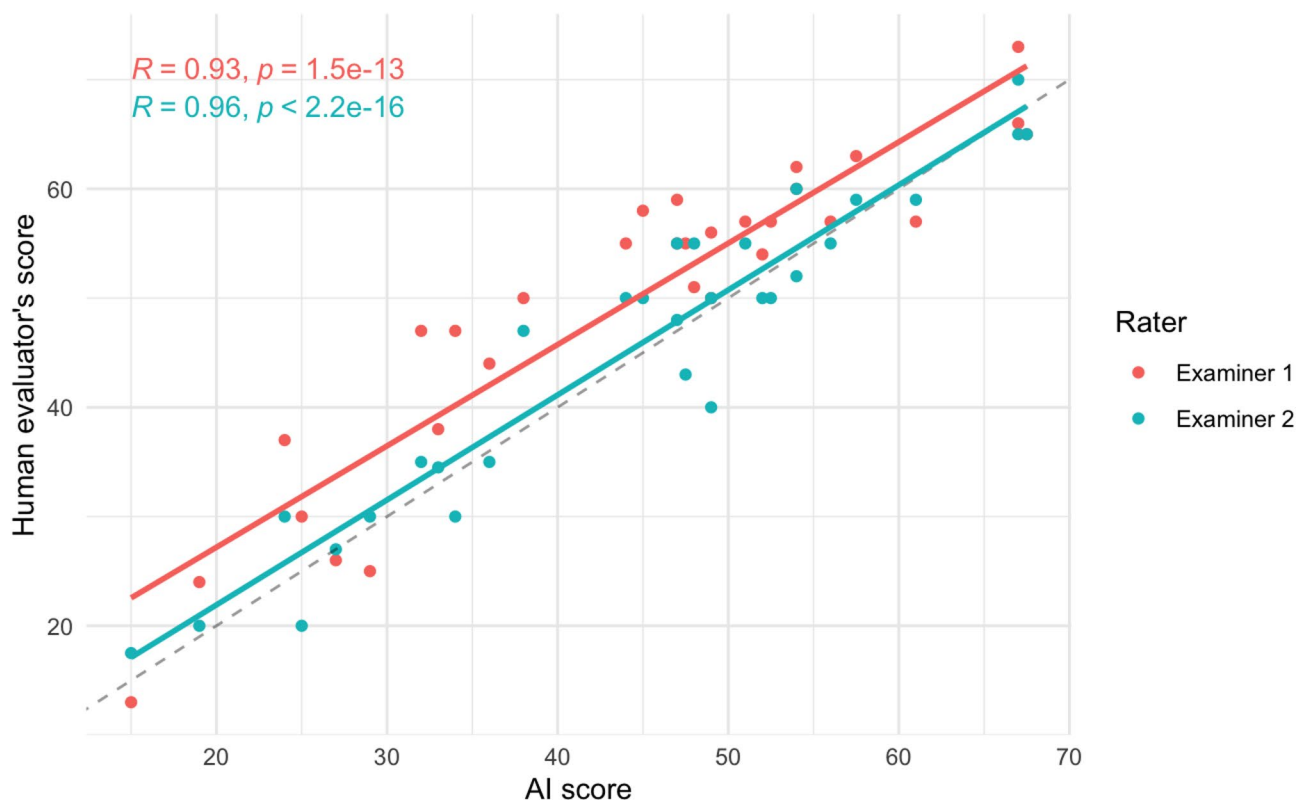


Fig. 4 A scatterplot depicting the association between AI score and the human evaluators' score. The two solid lines are the linear regression lines for each AI-human evaluator pair. The dashed line is the identity line

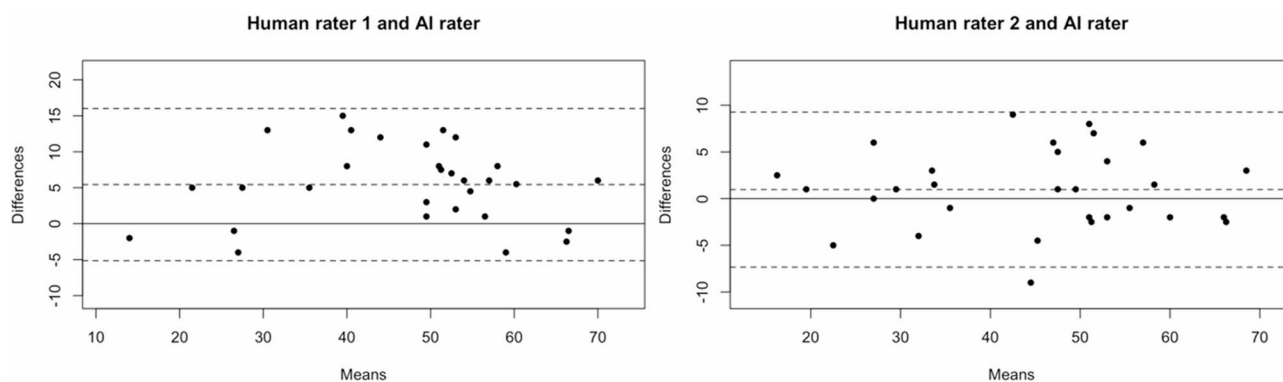


Fig. 5 Bland-Altman plots for the human raters' score and AI rater's score

sufficient general background knowledge from its training data.

However, some limitations should be noted. Although only a limited number of responses were evaluated, the post-hoc power of the study remains high, primarily due to the large observed effect size, which mitigates concerns associated with the small sample size.

While promising, more extensive evaluation would be needed to fully validate this approach across a wider range of disciplines and SAQ types. There may also be challenges in developing rubrics that are precise enough to ensure consistent AI scoring without ambiguity.

The proposed method tends to achieve better results when the topic is adequately represented in training data of the LLM. In other cases, external knowledge has to be made available through techniques like retrieval augmented generation. Therefore, the result presented here should not be generalized across domains.

Conclusion

AI assisted automated SAQ scoring tool markings highly correlate with the human examiner markings which shows promise for transparent, flexible automated grading with granular feedback. This approach

enabling criteria-based assessment could have valuable applications in education, reducing grading burdens on instructors while providing students with more granular, actionable feedback on their written work.

Future research could explore methods to further improve the rubric alignment process and evaluate the pedagogical impacts of deploying such AI-assisted essay scoring tools in real classroom environments. Ultimately, the synergy between advanced natural language AI and robust educational rubrics may open new possibilities for enhancing writing assessment and feedback.

Abbreviations

SAQ	Short answer questions
ASST	Automated SAQ scoring tool
AI	Artificial intelligence
LLMs	Large language models
NLP	Natural Language Processing
PEG	Project Essay Grade
IEA	Intelligent Essay Assessor
LSA	Latent semantic analysis
LSTM	Long Short-Term Memory
SVM	Support Vector Machine
ICC	Intra-class correlation coefficient

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12909-025-07718-2>.

Supplementary Material 1

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Author contributions

Seneviratne HMTW– Involved in developing the concept, design, data collection, analysis, drafting the manuscript, revising and final approval. Manatunga SS– Involved in design, analysis, drafting the manuscript, revising and final approval.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethical approval and consent to participate

Not applicable, since participation of human subjects was not entertained.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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